

Choosing a fraud model is a risk-acceptance decision, not a leaderboard

Detecting and estimating the financial cost of credit card fraud with machine learning: implications for risk management and regulatory compliance in UK financial institutions

Three models show no statistically significant difference in detection

Model	Test AUC	Precision	Recall	CV AUC
Logistic Regression	0.956	5.2%	87.4%	0.981
Random Forest	0.972	6.8%	85.3%	0.978
XGBoost	0.975	4.8%	87.4%	0.978

Paired t-tests on 5-fold CV AUC found no significant difference (only five folds, so limited power): LR vs RF $p = 0.29$, LR vs XGBoost $p = 0.43$, RF vs XGBoost $p = 0.92$.

At the default 0.5 threshold, every model raises about 14 to 20 false alarms for each fraud it catches.

2 · CALIBRATION

Calibration cuts probability error by 95% to 97%

Logistic Regression

95.0%

Brier score 0.0250 to 0.0012

Random Forest

96.8%

Brier score 0.0279 to 0.0009

XGBoost

96.7%

Brier score 0.0216 to 0.0007

Training on a balanced 378 + 378 sample distorts predicted probabilities. The Dal Pozzolo et al. (2015) correction restores them, which is what makes the cost estimate usable.

False alarms, not fraud, drive 72.7% of estimated loss

34,217.60 monetary units of expected loss on the test set (currency unconfirmed)



False alarms: 24,876.77 (72.7%) Fraud caught: 8,926.22 (26.1%) Unflagged: 414.62 (1.2%)

One genuine transaction of 25,691.16, flagged at 68.6%, accounts for 17,622.60 of the total: 51.5%.

Explanations are stable but weakly fitted, so compliance is only partly evidenced

Provision	Finding
Consumer Duty: foreseeable harm	Well supported: false alarms are 72.7% of estimated loss
DUAA 22C(a): information	Satisfiable, but top features are anonymised components
DUAA 22C(d): ability to contest	Partial: SHAP stable (V14 top in 5/5 RF folds), LIME Jaccard 0.69 to 0.77 but R^2 only 0.37 to 0.39
PRA SS1/23 Principle 3: documentation	Satisfied in principle
PRA SS1/23 Principle 4: validation	Tension: XGBoost has the best AUC and is hardest to validate

Six further provisions depend on firm-level process, so model output alone cannot test them.

Treat model choice as a documented risk-acceptance decision

Record the trade-off

Weaker detection with an easier compliance case (LR), or stronger detection with a harder one (RF, XGBoost).

Monitor concentration

Track false-positive concentration, not just count: one transaction held 51.5% of estimated loss.

Audit explanations twice

Check stability across reruns and local fit (R^2) before relying on LIME for compliance.

Price in validation

Count SS1/23 documentation and validation effort as a quantified input to model selection.